

## Exercise Solution

## problem 4.2: Bayesian Sensor Reasoning

From [Problems](#)

### Setup

- Prior:  $P(D = 1) = 0.15$ , so  $P(D = 0) = 0.85$
- Old sensor: true-positive rate  $P(C = 1 \mid D = 1) = 0.85$ , false-positive rate  $P(C = 1 \mid D = 0) = 0.08$

### Part 1: No Creaking Observed

We want  $P(D = 1 \mid C = 0)$ .Prior:  $P(D = 1) = 0.15$ Likelihood (for  $C = 0$ ):

$$P(C = 0 \mid D = 1) = 1 - 0.85 = 0.15 \quad (\text{false negative: damage present but no creaking})$$

$$P(C = 0 \mid D = 0) = 1 - 0.08 = 0.92 \quad (\text{true negative: no damage, no creaking})$$

Evidence (law of total probability):

$$\begin{aligned} P(C = 0) &= P(C = 0 \mid D = 1) P(D = 1) + P(C = 0 \mid D = 0) P(D = 0) \\ &= 0.15 \times 0.15 + 0.92 \times 0.85 = 0.0225 + 0.782 = 0.8045 \end{aligned}$$

Posterior (Bayes' rule):

$$P(D = 1 \mid C = 0) = \frac{P(C = 0 \mid D = 1) P(D = 1)}{P(C = 0)} = \frac{0.15 \times 0.15}{0.8045} = \frac{0.0225}{0.8045} \approx 0.0280$$

No creaking reduces the probability of damage from 15% to about 2.8%.

### Part 2: Creaking Observed

We want  $P(D = 1 \mid C = 1)$ .Likelihood (for  $C = 1$ ):

$$P(C = 1 \mid D = 1) = 0.85, \quad P(C = 1 \mid D = 0) = 0.08$$

Evidence:

$$P(C = 1) = 0.85 \times 0.15 + 0.08 \times 0.85 = 0.1275 + 0.068 = 0.1955$$

Posterior:

$$P(D = 1 \mid C = 1) = \frac{0.85 \times 0.15}{0.1955} = \frac{0.1275}{0.1955} \approx 0.6522$$

Hearing creaking raises the probability from 15% to about 65%.

**Check:** The two posteriors should be consistent with the evidence probabilities:

$$\begin{aligned} P(D = 1 \mid C = 0) \times P(C = 0) + P(D = 1 \mid C = 1) \times P(C = 1) &= P(D = 1) \\ 0.0280 \times 0.8045 + 0.6522 \times 0.1955 &\approx 0.0225 + 0.1275 = 0.15 \checkmark \end{aligned}$$

### Part 3: Sensor Comparison

New sensor:  $P(C = 1 \mid D = 1) = 0.95$ ,  $P(C = 1 \mid D = 0) = 0.02$ .

**No creaking ( $C = 0$ ):**

$$\begin{aligned} P(C = 0 \mid D = 1) &= 0.05, & P(C = 0 \mid D = 0) &= 0.98 \\ P(C = 0) &= 0.05 \times 0.15 + 0.98 \times 0.85 = 0.0075 + 0.833 = 0.8405 \\ P(D = 1 \mid C = 0) &= \frac{0.05 \times 0.15}{0.8405} = \frac{0.0075}{0.8405} \approx 0.0089 \end{aligned}$$

**Creaking ( $C = 1$ ):**

$$\begin{aligned} P(C = 1) &= 0.95 \times 0.15 + 0.02 \times 0.85 = 0.1425 + 0.017 = 0.1595 \\ P(D = 1 \mid C = 1) &= \frac{0.95 \times 0.15}{0.1595} = \frac{0.1425}{0.1595} \approx 0.8934 \end{aligned}$$

**Comparison summary:**

|                       | Old sensor | New sensor |
|-----------------------|------------|------------|
| $P(D = 1 \mid C = 0)$ | 0.028      | 0.009      |
| $P(D = 1 \mid C = 1)$ | 0.652      | 0.893      |

The better sensor is more informative in both directions: silence is stronger evidence of safety (0.9% vs. 2.8%), and creaking is stronger evidence of damage (89% vs. 65%). The lower false-positive rate means creaking is more trustworthy; the higher true-positive rate means silence is more exonerating.

### Part 4: Multiple Adjacent Squares

Define:

- $D_1$ : damage at  $(2, 1)$ ,  $D_2$ : damage at  $(1, 2)$ , both with prior 0.15
- $D_1, D_2$  independent
- $A$ : at least one adjacent square is damaged

**Step 1:** Probability of at least one damaged square.

$$P(A) = 1 - P(D_1 = 0) P(D_2 = 0) = 1 - 0.85 \times 0.85 = 1 - 0.7225 = 0.2775$$

So  $P(\text{none damaged}) = 0.7225$ .

**Step 2:** Compute  $P(C = 0)$  via total probability.

$$\begin{aligned} P(C = 0) &= P(C = 0 \mid A) P(A) + P(C = 0 \mid \bar{A}) P(\bar{A}) \\ &= (1 - 0.85) \times 0.2775 + (1 - 0.08) \times 0.7225 \\ &= 0.15 \times 0.2775 + 0.92 \times 0.7225 \end{aligned}$$

$$= 0.041625 + 0.6647 = 0.7063$$

**Step 3:** Compute  $P(D_1 = 1 \mid C = 0)$ .

We need  $P(D_1 = 1, C = 0)$ , which requires summing over  $D_2$ :

$$P(D_1 = 1, C = 0) = \sum_{d_2} P(C = 0 \mid D_1 = 1, D_2 = d_2) P(D_1 = 1) P(D_2 = d_2)$$

When  $D_1 = 1$ , at least one square is damaged regardless of  $D_2$ , so  $P(C = 0 \mid D_1 = 1, D_2 = d_2) = 1 - 0.85 = 0.15$  for both values of  $d_2$ :

$$P(D_1 = 1, C = 0) = 0.15 \times 0.15 \times 0.85 + 0.15 \times 0.15 \times 0.15 = 0.15 \times 0.15 \times (0.85 + 0.15) = 0.15 \times 0.15 = 0.0225$$

Therefore:

$$P(D_1 = 1 \mid C = 0) = \frac{0.0225}{0.7063} \approx 0.0319$$

This is slightly higher than the single-square result from Part 1 ( $\approx 0.028$ ). Why? The numerator

$P(D_1 = 1, C = 0) = 0.0225$  is identical in both cases. The difference is entirely in the denominator:

$P(C = 0) = 0.7063$  here versus  $0.8045$  in Part 1. With two potential damage sources, no-creaking is a rarer event overall (there are more ways for at least one square to be damaged and trigger the sensor). Since no-creaking is less common, observing it is less informative about any *single* square—it doesn't exonerate **(2, 1)** as strongly.

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